GEORGE MASON UNIVERSITY

CHALLENGE X

Use Case Report

Project Title:

Agentic AI based Property Valuation

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Problem Overview

- Improve efficiencies in the designation process of a property due to fragmented property and market data sources. Manual collection and reconciliation of valuation reports, property imagery, and market insights is time-consuming and prone to error. Creating a unified, interface can streamline property valuation process and maintain historical data for audit.

Note: This is an academic engagement with the goal of developing students' technical and problem-solving skills. No Fannie Mae specific data, tools, facilities or processes will be used.

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Challenge

- Read and scan valuation product files in PDF/XML formats from public data sources like County and Local Property Appraiser websites, Open Data Portals, FHFA and HUD, and Freemium Platforms like Zillow, Redfin, CoreLogic, House Canary, Red Bell etc.
- Apply computer vision to analyze property and comparable images.
- Coalesce data across products, databases and external sources.

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Proposed Solution

Build a chatbot and/or user interface that:

- Flag all anomalies into products for analyst review.
- Key datapoint alignment (example; appraisal, broker price opinion, computer-vision returned property condition from vendors such as House Canary, RedBell, MLS, Zillow etc.)
- API to AVM models and values, provide floor and ceiling values, and range for guidance
- Enable drill-down capability via chatbot for property, comparable, or market questions

- Security & Compliance

- Responsible, Safe & Secured use of AI
- Supports secure deployment and future integration
- Sensitive data handling
- Data retention and usage within model ecosystem
- Data integrity, accuracy and repeatability

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Impact & Goals

Improved Response Times: KPIs: Improve response times for property valuation request. Zero degradation of performance in aggregate.

Accelerated Time-to Market: Centralizes data for faster, more accurate designation decisions. Records all interactions and stores them as part of the historical property roadmap

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Prizes

1st Place

Vert Backpack

Mobile Office Commuter Sleeve

New! Stanley AeroLight 24oz Bottle

2nd Place

Vert Backpack

Mobile Office Commuter Sleeve

3rd Place

Vert Backpack